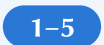


## MAVRA middleware



learning



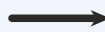
find definition



run SQL



call / response



write to memory



read database



### Built-in agent (LLM)

learns definitions, proposes rewrites

admit, publish

### Query service

#### Find definition

match name, check data

#### Pre-run check

validate on one snapshot

### Shared memory

also table profiles, join paths

#### Definitions

revenue v2  
pending

electr. v2  
valid

rate v2  
valid

#### Validation cache

per (rule, data version), shared

one row/sale  
version v15 ✓

loss in bound  
version v15 ?

date key unique  
version v3 ✓

3 missing: run check

4 store

new revision or retire

### Maintenance

checks · repair, optimize, retire

check SQL, data versions

run SQL on the snapshot

## User agents

Store revenue  
in September?

SQL

metrics: revenue v2

1 ask "revenue"

5 valid v2

6 run SQL

8 result

## ETL / writers

write

### Database

snapshot

store\_sales v14 → v15  
date\_dim v3

store\_sales

date\_dim